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Software engineer since 2018. Builds automatic graders (the heuristic guide's checker, acu-primer's answer grading) and works daily with coding agents (Claude Code, Codex). IMO 2015 (Uganda); Codeforces max rating 1661.

Projects

- **acu-primer**: Graded lesson player and authoring tool (work in progress; live demo linked; Devpost Build With AI: Basics entry)
  - **Each check is graded against a rubric of anticipated wrong answers**: The wrong answers are exact options or numeric ranges, each tied to a misconception from a catalogue with cited research, plus a general fallback for unplanned answers and a two-level stop. Lessons are drafted from textbook pages by a **Claude Code skill** (`/draft-lesson`) and must pass the app's own schema and consistency checks (for example, no wrong-answer range may overlap the right answer) before import. A person signs a lesson off before students see it. 43 unit tests. TypeScript, Svelte 5, Cloudflare Workers + D1.
- **Heuristic guide**: Ten-level course for heuristic optimization contests (AtCoder AHC, CodinGame, ICPC Challenge)
  - **Each level ends in an automatic grader**: `python -m lab.check L2 solver.py` re-runs a baseline on the same seeds and time limit, and passes the solver only if it clears a threshold calibrated against a reference solution *and* the paired difference beats its own noise ($|t| > 2$). Four problem types (TSP, scheduling, packing, tiles), a race harness and a parameter sweep tool; thresholds calibrated for levels 0 to 3 so far. Standard-library Python.
- **AcuGemma**: On-device AI tutoring app
  - **Offline tutor for primary school students**: Runs the Gemma 3n 2B model on an Android phone through MediaPipe, so it works fully offline. A prompting system guides each lesson, evaluates the student's answers and gives feedback in context.
- **Uganda Coffee Traceability Dashboard**: Geospatial verification tool
  - **Checks 97,000+ coffee plots against 2020 satellite baselines**: 10m-resolution Google Earth Engine data in a GeoDjango and PostGIS backend, a GraphQL API for coordinate checks, and a React-Leaflet interface with district filtering for dense spatial layers.
- **Writing**: Monotonic deques and stacks, Part 1 (Oct 2026, with an interactive algorithm stepper) and Learning cryptography with an AI guide: Cryptopals Set 1 (Aug 2026, Claude Code as critic and tutor)

Competitions

- **International Mathematical Olympiad (IMO) 2015, Chiang Mai, Thailand**: Represented **Uganda** on the national team.
- **ICPC 2026 Online Challenge 1, powered by Huawei (Codeforces, Aug 2026)**: Final rank **1,173 of 4,428** after system tests (13,593.7 points; 338th on the provisional pretest standings). Interactive scheduling for edge-cloud LLM inference. Wrote a C++ solver, plus a local interactor and scorer that matched the judge's worked example on the first run, seeded test generators and rehearsal suites, built with Claude Code as coding partner.
- **CodinGame Summer Challenge 2026 (Sep 2026)**: **Silver league, 124 of 628** in Silver. Python bot iterated over versions against the official referee run locally.
- **Competitive programming**: Codeforces (max rating 1661), AtCoder (max rating 1256) and Project Euler; solutions and algorithm library in competitive-programming.

Experience

- **Tunga** — Kampala, Uganda
  Backend Software Engineer (Django) — *Nov 2023 - Present*
  - **AI talent matching and outreach agent (tunga-agent)**: Led development of a multi-stage agentic pipeline that classifies companies, matches developers to open roles, and runs multi-variant email and LinkedIn outreach with engagement-triggered follow-ups. Matching uses **RAG** over Pinecone (OpenAI embeddings, top-K cosine matching with thresholds and a relaxed fallback), and **GPT-4o-mini** classifies company fit and writes personalized outreach. Cut LLM costs ∼**90%** with hybrid OpenAI/local **Ollama** inference, domain-keyed classification caching and hard pre-filters. Exposed a **Model Context Protocol (MCP)** server with live pipeline analytics and tools for Claude-driven lead collection. Instrumented with Sentry, health probes and a full audit trail; deployed on Kubernetes with Celery/Redis.

- **AI-powered B2B lead identification (Lead-ID)**: Designed a lead-scoring engine that detects **40+ buying-intent signal types across 6 intent streams**, with recency-decay weighting, per-signal caps and multi-stream reinforcement. Generalized it to **three verticals** (IT staffing, accounting & finance, AI consulting) through per-use-case rules, weights and thresholds stored in the database, so tuning needs no code change. GPT-4o-mini classifies job descriptions and writes "why now" narratives, with a regex fallback when the LLM is unavailable. Processes **500+ companies per batch** and tracks score changes across runs.
- **Coding agents in daily work**: Build and review with **Claude Code** (project skills, MCP connectors) and Codex. Review teammates' merge requests against written step-by-step test checklists.
- **Tunga Desk**: Built an Electron prototype that runs Claude Code or Codex headless over the team's shared AI kit (skills, team agents, a daily decision log), as part of Tunga's AI-first initiative.
- **Tunga Works platform**: Built features across Tunga Works, the platform where developers, clients and staff run their engagements: timesheets, invoicing, payouts and onboarding. Ran a performance audit of the backend and frontend, and profiled and sped up slow endpoints (N+1 queries, lighter serializers for search). Features include invoicing (line items, multi-developer and partial invoices, credit notes, Exact Online sync), USD and Payoneer payouts, a timesheets redesign with comment threads and notifications, and onboarding with an AI resume parser.

- **Sunbird AI** — Kampala, Uganda
  Software Engineer and Team Lead — *May 2020 - Oct 2023*
  - **Leadership**: Led and mentored a team of developers and interns that took ML models to production and ran the MLOps workflows.
  - **Model deployment and API**: Deployed the language translation models to an Amazon SageMaker endpoint, then led the migration to GCP Vertex AI. Built and led the Sunbird AI API (translation, speech recognition and text-to-speech in Ugandan languages), used by Ugandan companies in their apps and in the IndabaX language technology hackathon.
  - **Translation frontend**: Rewrote the Ugandan languages translation app from vanilla JavaScript to React, which made new features faster to build.
  - **Noise monitoring**: Built the back end and front end of the environmental noise monitoring system in Kampala and Entebbe, then moved its core from a self-hosted setup to AWS IoT. The Mayor of Entebbe uses it to monitor and reduce noise in the city.
- **Andela** — Kampala, Uganda
  Software Engineer — *Aug. 2018 – Apr. 2020*
  - **Android development**: Built offline features for Converge (an internal meeting-room app) and instrumented tests that simulate going offline, using Test Butler. Mentored apprentice developers through tasks and code reviews.
- **Makerere University Data Science and Artificial Intelligence Lab** — Kampala, Uganda
  Software Engineer Intern — *Jun. 2017 – Sept. 2017*
  - **Android development**: Built photo annotation in the White-Fly count app, so farmers could mark white-fly infested cassava leaves.

Education

- **Makerere University** — Kampala, Uganda
  Bachelor of Science in Computer Science — *Aug. 2015 - Apr. 2018*
- **Coursera**: Machine Learning Specialization, Mathematics for Machine Learning, Deep Learning Specialization

Skills

- **Languages**: Python, TypeScript/JavaScript, C++, Kotlin
- **AI tools**: Claude Code (skills, MCP servers), Codex, OpenAI API, Ollama, Pinecone, Gemma via MediaPipe
- **Technologies**: Django, FastAPI, React, Svelte, PostgreSQL, Celery/Redis, Kubernetes, Cloudflare Workers + D1, AWS, GCP